Problem Set #4

Kabir Kanha Arora
ECO 354 - Public Economics
Shiv Nadar University

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Question 1

Assume that individuals have the same utility function over consumption and labor given by $U(c,l) = (1-\theta)lnc + \theta ln(50-l)$ where c represents consumption and l represents hours of labor and $\theta \in (0,1)$ is a given parameter. Here, $ln(x)$ denotes the natural logarithm of x (this can also be denoted by $log(x)$). Assume also that the only income that individuals have is from labor income and that the hourly wage rate is given by w.

(a) Write the budget constraint faced by the individual.

(b) Set up the maximization problem of this individual and solve for the optimal choices of labor and consumption.

(c) Now, suppose that there is a tax of $\tau = 20\%$ is imposed on labor income. Solve for the new optimal choice of labor and consumption.

(d) On a graph, plot the solutions you found in parts (b) and (c). Be sure to label all intercepts, optima, as well as the income and substitution effects.

(e) For the individual's choice of labor, explain intuitively the direction of the substitution and income effect of the introduction of the tax. Does the income effect or substitution effect dominate? Explain your findings.

(a) The individual has no endowment, the only income they get is their labour income. Clearly, the total amount of income earned from labour is wl. Hence, since net income has to be at least as much as the consumption. So their budget constraint is:

$$c \leq wl$$

(b)

$$max_{c,l} U(c,l) = (1-\theta)lnc + \theta ln(50-l)$$

Clearly, utility will be maximised when

$$c = wl$$

Plugging this into the objective function, we get

$$max_l U(c,l) = (1-\theta)ln(wl) + \theta ln(50-l)$$

Differentiating U w.r.t l:

$$\frac{dU}{dl} = \frac{(1-\theta)w}{wl} + \frac{\theta(-1)}{50-l}$$

First Order Condition:

$$\frac{(1-\theta)}{l} = \frac{\theta}{50-l}$$

$$\implies l\theta = (1-\theta)(50-l)$$

$$\implies l\theta = 50 - l - 50\theta + l\theta$$

$$\implies \boldsymbol{l^* = 50(1-\theta)}$$

We know, $c = wl$, so:

$$\implies \boldsymbol{c^* = 50w(1-\theta)}$$

(c) $\tau = 20\%$. So, $c = wl(1-\tau)$, $\implies c = 0.8wl$

Plugging this into the objective function, we get

$$max_l U(c,l) = (1-\theta)ln(0.8wl) + \theta ln(50-l)$$

Differentiating U w.r.t l:

$$\frac{dU}{dl} = \frac{(1-\theta)0.8w}{0.8wl} + \frac{\theta(-1)}{50-l}$$

First Order Condition:

$$\frac{(1-\theta)}{l} = \frac{\theta}{50-l}$$

$$\implies l\theta = (1-\theta)(50-l)$$

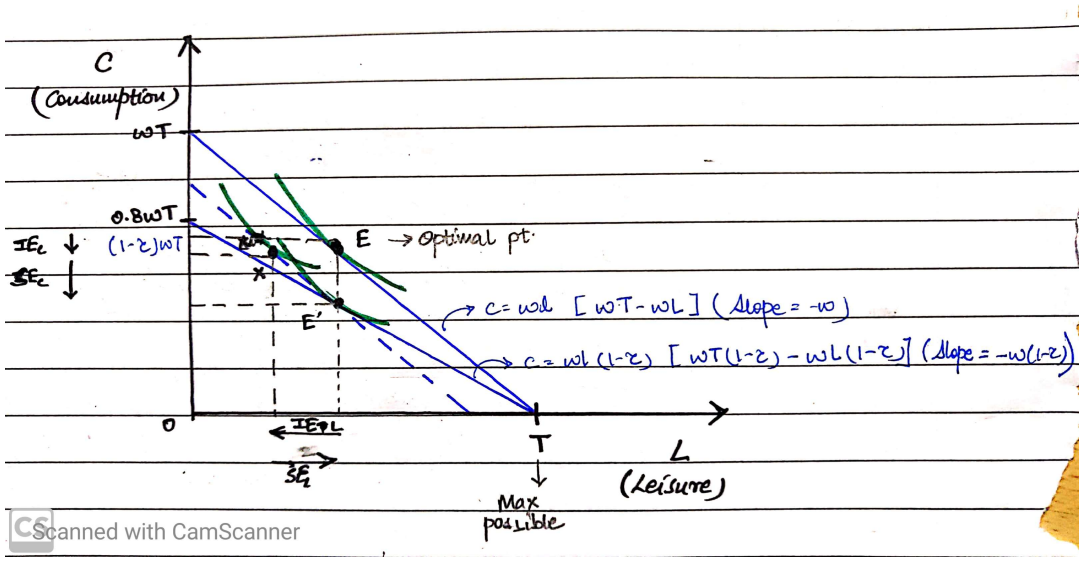
$$\implies l\theta = 50 - l - 50\theta + l\theta$$

$$\implies \boldsymbol{l^* = 50(1-\theta)}$$

We know, $c = 0.8wl$, so:

$$\implies \boldsymbol{c^* = 40w(1-\theta)}$$

(d) The graph is shown below. **E** refers to the point found in (b) and **E'** refers to the point found in (c). IE_L, SE_L, IE_c, SE_c refer to the income and substitution effects for leisure and consumption respectively.



(e) We know that when tax increases, the cost of leisure goes down by a factor of $(1-\tau)$. So, by **substitution effect**, L goes up and c goes down. However, since effective income has gone down, and both goods are normal, both c and L go down by **income effect**, where L denotes the hours of leisure and is calculated as $T - l$, where T is the total number of hours in the time period.

We see that both income and substitution effects push the consumption **down**, and hence the effect is reinforced and clearly it falls. But for leisure, substitution effect is pushes it up while income effect pushes it down. Here, we see that labour supply does not depend on wage rate (inelastic) and hence the substitution and income effects cleanly **cancel out** each other, and the labour supply remains constant. These results are also explained in the graph.

Question 2

Consider a forced savings policy and how it affects two types of people - patient and myopic. Both types have the following problem:

$$max_{c1,c2} \ln c_1 + \beta \ln c_2$$

such that,

$$c_1 + s = 100$$
$$c_2 = s(1+r)$$

where c_1 is consumption in period 1, β is the discount rate, c_2 is consumption in period 2, s is savings, r is the interest rate, and $\ln c$ [logarithm of c] denotes the individual's utility from consuming c in a given period.

(a) **What is the optimal ratio of consumption in period one to consumption in period two? Explain this relationship intuitively.**

(b) Suppose that the interest rate is 20% for the remainder of the problem. What is the optimal amount of savings s for patient people who have $\beta = 1$?

(c) What is the optimal amount of savings s for myopic people who have $\beta = 0$?

Suppose that the government imposes a lump sum tax of $\tau = 50$ to be paid in period 1 and pays benefits of $b = 50(1 + r)$ back in period 2. Note that people cannot borrow against these period 2 benefits.

(d) How much does the patient type save? Does this plan make them better off in period 2? Does this plan make them better off overall?

(e) How much does the myopic type save? Does this plan make them better off in period 2? Does this plan make them better off overall?

(a) The optimal consumption relationship for the two periods take into account both the interest rate r and the discount rate β. While the future consumption is valued higher because of r, it is valued lower because of β. The relationship is given by

$$\frac{1}{c_1} = \frac{\beta(1+r)}{c_2}$$

This balance means that on the margin people are indifferent between consuming in period 1 and period 2. The fractions indicate that giving up one unit of utility today gives higher utility in the next period by a factor of $(1 + r)$, but it is discounted by a factor of β, which is less than or equal to 1.

Derivation:

Utility function:

$$U = \ln c_1 + \beta \ln c_2$$

This can be re-written as:

$$U = \ln(100 - s) + \beta \ln(s(1 + r))$$

Differentiating w.r.t s, we get:

$$\frac{dU}{ds} = \frac{-1}{100 - s} + \frac{\beta(1+r)}{s(1+r)}$$

First order condition:

$$\frac{1}{100 - s} = \frac{\beta(1+r)}{s(1+r)}$$

Which, from the given equations, is essentially the same as:

$$\boldsymbol{\frac{1}{c_1} = \frac{\beta(1+r)}{c_2}}$$

(b) We are given that

$$\beta = 1$$

$$r = 0.2$$

Substituting this in the equating generated in the previous part, we get,

$$\frac{1}{c_1} = \frac{1.2}{c_2}$$

$$\implies c_2 = 1.2c_1$$

Using this in the equation

$$c_2 = s(1 + r)$$

We get

$$1.2c_1 = 1.2s$$

$$\implies c_1 = s$$

Substituting this in the equation

$$c_1 + s = 100$$

We get

$$2s = 100$$

$$\implies s = 50$$

Hence, the optimal amount of savings for the patient person is **50 units**.

(c) This part is fairly intuitive, if $\beta = 0$, then it means that the consumption in period 2 has no utility for the myopic person. Hence, the optimal amount of savings is clearly **0**.

(d) Since the lump sum tax τ gives the same rate of interest r, it can be thought of as a compulsory saving that everyone has to make. The new savings s is now equal to $s' + \tau$. Nothing else has changed and since we already established that the optimal saving amount for a patient person is 50, they will not make any additional explicit savings since 50 is already being saved compulsorily. Hence, $\boldsymbol{s' = 0}$. This does not make them better off or worse and they are in the same position as they were before the tax.

(e) The myopic person has no utility for what they consume in period 2, so he they do not make any additional savings and consume the entire 50 units in the first period. In the second period, they now get $50 \times 1.2 = 60$ units as compared to 0 earlier, so they are definitely better off in period 2, however since $\beta = 0$, from the utility function, it is clear that their utility has gone down from $ln(100)$ to $ln(50)$, and hence they are definitely worse off overall.